UNIVERSITY OF CALGARY

ENEL 500 – Capstone

Final Design Report

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Objectives

This is a comprehensive report that covers all the details of the project.

Document Structure

The design final report must include the following sections.

1. Project Identification Cover Page

Create a cover page which includes the following information (same as previous report):

- Project title
- Team number
- Name, email, UCID of the project manager at the time
- Name and UCID for the rest of the team members
- Name of the support technician if applicable
- Name of the academic advisor
- Name of the sponsoring company
- Name and email of the sponsor representative

2. Table of Content

Ensure the document has a table of contents.

3. Glossary

In this section, include terms, abbreviations, and nomenclature that would help with the readability of the report.

4. Executive Summary

This section provides a brief overview of the project and briefly mentions the main points. You may look at this section from some industry reports as guidelines for how to write an executive summary. By reading this part, the reader should gain a good understanding of the main objectives, motivations, and accomplishments of the project.

5. Project Motivation and Objectives

Describe the project motivations and objectives. Be specific in terms of issues that might have led to the definition of this project (motivation), and what this project aimed to accomplish (objectives).

6. Methodology and Design

In this section, a detailed description of the final methodology and the design for creating the final product needs to be provided. This is where you present the details of your design and the methodology that you have implemented. You need to include supplementary materials, such as graphs, flowcharts, circuit diagrams, and any other necessary work detail to make the methodology and design details clear. Imagine that someone wants to duplicate this project. What you present here should ideally be enough to duplicate the project.

7. Product Scope and Final Product Functionality

In this section, elaborate on the scope of your final product and what it does and does not do. If you have deviated from the original scope promised in the Fall term, identify and justify why.

8. Technical Specifications of the Final Product

Describe the final technical specifications of the final product. Identify and justify any deviations from the Fall term specifications.

9. Measuring Success and Validation Test Results

You need to elaborate here on why the project is successful, based on measures that you had identified before. You may add new measures as well. Also, to demonstrate that the final products meet the expectations of the sponsor as defined in the Fall term, you need to identify and demonstrate relevant validation tests. Present the details of the validation tests and discuss how the tests prove the soundness of the final product.

10. List of Tools, Materials, Supplies, Costs, etc.

Present a finalized list of tools, materials, services, and supplies, whichever is applicable, along with the associated costs, that were used to make the project successful. Compare with the ones that you envisioned in the Fall term report and comment.

11. Reflection on Ethics and Equity in your Capstone Journey

Reflect on ethical considerations and equity issues encountered during your project. Discuss how your team identified and addressed ethical dilemmas, ensuring adherence to professional codes of conduct. Reflect on how equity considerations influenced your design decisions and team interactions.

12. Reflection on Impact of Engineering on Society/Environment

Reflect on the societal and environmental implications of your project. Discuss both positive and negative impacts, and describe measures taken to enhance benefits and mitigate adverse effects.

13. Reflection on Team’s Approach to Project Management

Reflect on your team's project management strategies, including planning, scheduling, resource allocation, and risk management. Discuss how these approaches facilitated the project's progression, and any challenges encountered.

14. Reflection Team's Approach to Economics and Cost Control

Reflect on the economic aspects of your project, including budgeting, cost estimation, and financial decision-making processes. Discuss how economic considerations influenced design choices and project scope.

15. Reflections on Team Member's Approach to Life-long Learning

Reflect on how team members identified knowledge gaps and sought new information or skills to overcome challenges during the project. Discuss the strategies employed for continuous learning and how this experience has prepared you for ongoing professional development.

16. Appendix

Include supplementary materials like raw data, calculations, drawings, and charts. Label and reference each item in the main report.

17. References (if any)

Cite all the sources that were used in this report in this section.

Formatting Guidelines

- Formatting requirements for the report body materials is flexible and is left to your choice
- Font: Times New Roman, 12 pt
- Spacing: 1.5 lines
- Margins: Normal 1''

Submission Requirements

- Submit through D2L only. Detailed instructions will be provided closer to the submission deadline.
- One submission per team

- File format: PDF only
- Grading rubric is available well before the deadline for clarity and preparation
- https://d2l.ucalgary.ca/d2l/le/content/618468/viewContent/6852219/View